Physics

Time Allowed : 2 hour ___________________________ **Maximum Marks : 180**

Please read the instructions carefully. You will be alloted 5 minutes specifically for this purpose.

Instructions

A. General

1. Blank papers, clipboards, log tables, slide rules, calculators, cellular phones, pagers, and electronic gadgets in any form are not allowed.
2. Do not break the seals of the question-paper booklet before instructed to do so by the invigilators.

B. Question paper format and Marking Scheme :

1. This question paper contains 10 questions. Each question in this section contains 15 marks.

Q1: Attempt to formulate your 'moral' views on the practice of science. Imagine yourself stumbling upon a discovery, which has great academic interest but is certain to have nothing but dangerous consequences for the human society. How, if at all, will you resolve your dilemma ?

Q2: A man wishes to estimate the distance of a nearby tower from him. He stands at a point A in front of the tower C and spots a very distant object O in line with AC. He then walks perpendicular to AC up to B, a distance of 100 m, and looks at O and C again. Since O is very distant, the direction BO is practically the same as AO; but he finds the line of sight of C shifted from the original line of sight by an angle $\vartheta = 40^0$ (ϑ is known as 'parallax') estimate the distance of the tower C from his original position A.

Q3: A player throws a ball upwards with an initial speed of 29.4 m s^{-1}.

(a) What is the direction of acceleration during the upward motion of the ball ?

(b) What are the velocity and acceleration of the ball at the highest point of its motion ?

(c) Choose the x = 0 m and t = 0 s to be the location and time of the ball at its highest point, vertically downward direction to be the positive direction of x-axis, and give the signs of position, velocity and acceleration of the ball during its upward, and downward motion.

(d) To what height does the ball rise and after how long does the ball return to the player's hands ? (Take g = 9.8 m s–2 and neglect air resistance).

Q4: It is now believed that protons and neutrons (which constitute nuclei of ordinary matter) are themselves built out of more elementary units called quarks. A proton and a neutron consist of three quarks each. Two types of quarks, the so called 'up' quark (denoted by u) of charge + (2/3) e, and the 'down' quark (denoted by d) of charge (–1/3) e, together with electrons build up ordinary matter. (Quarks of other types have also been found which give rise to different unusual varieties of matter.) Suggest a possible quark composition of a proton and neutron.

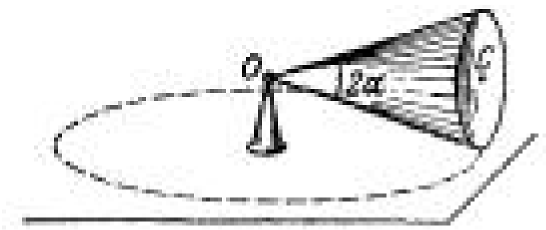
Q5: In a Van de Graaff type generator a spherical metal shell is to be a 15×10^6 V electrode. The dielectric strength of the gas surrounding the electrode is 5×10^7 Vm–1. What is the minimum radius of the spherical shell required?

Q6: A balloon starts rising from the surface of the Earth. The ascension rate is constant and equal tov_0. Due to the wind the balloon gathers the horizontal velocity component v_x = ay, where a is a constant and y is the height of ascent. Find how the followlllg quantities depend on the height of ascent:

(a) the horizontal drift of the balloon x (y);

(b) the total, tangential, and normal accelerations of the balloon.

Q7: A solid body rotates with a constant angular velocity w_o = 0.50 rad/s about a horizontal axis AB. At the moment t=0 the axis AB starts turning about the vertical with a constant angular acceleration β_o = 0.10 rad/s^2. Find the angular velocity and angular acceleration of the body after t = 3.5 s.

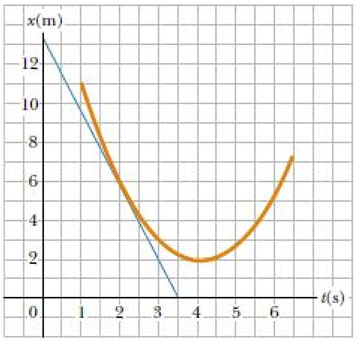


Q8: A position–time graph for a particle moving along the x axis is shown in Figure .

(a) Find the average velocity in the time interval $t = 1.5s$ to $t = 4.0s$

(b) Determine the instantaneous velocity at $t = 2.0s$ by measuring the slope of the tangent line shown in the graph.

(c) At what value of t is the velocity zero?



Q9: *Every morning at seven o'clock*

There's twenty terriers drilling on the rock.

The boss comes around and he says, "Keep still

And bear down heavy on the cast-iron drill

And drill, ye terriers, drill." And drill, ye terriers, drill.

It's work all day for sugar in your tea . . . And drill, ye terriers, drill.

One day a premature blast went off

And a mile in the air went big Jim Goff. And drill . . .

Then when next payday came around

Jim Goff a dollar short was found. When he asked what for, came this reply:

"You were docked for the time you were up in the sky." And drill . . .

—American folksong

What was Goff's hourly wage? State the assumptions you make in computing it.

Q10: A sphere of radius r carries a surface charge of density$\sigma = \vec{a}.\vec{r}$, where a is a constant vector, and r is the radius vector of a point of the sphere relative to its centre. Find the electric field strength vector at the centre of the sphere.

Q11: An electric capacitor consists of thin round parallel plates, each of radius R, separated by a distance l (l«R) and uniformly charged with surface densities σ and $-\sigma$. Find the potential of the electric field and the magnitude of its strength vector at the axes of the capacitor as functions of a distance x from the plates if $x \gg l$. Investigate the obtained expressions at $x \gg R$.

Q12: A uniformly distributed space charge fills up the space between two large parallel plates separated by a distance d. The potential difference between the plates is equal to $\triangle\phi$. At what value of charge density ρ is the field strength in the vicinity of one of the plates equal to zero? What will then be the field strength near the other plate?